

Probabilistic Day-Ahead Electricity Price Forecasting for Germany–Luxembourg and Spain

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Abstract

We forecast hourly day-ahead electricity prices for the German–Luxembourg (DE-LU) and Spanish (ES) bidding zones with calibrated probabilistic intervals. The pipeline combines a gradient-boosted quantile regression (short-term, ≤ 14 days) with an STL-based seasonal extrapolation (long-term, > 14 days), trained on price, generation, weather, and fuel data from 2024-01-01 onwards. Held-out backtesting on Sunday 2026-05-10 yields a pinball loss at $q = 0.45$ of 12.59 for DE-LU and 14.86 for ES, with 95% prediction-interval coverage of 90.9% and 63.6% respectively. We discuss the systematic ES overnight bias on Sundays and outline targeted recalibration.

1 Problem statement

Let $p_t^z \in \mathbb{R}$ denote the day-ahead auction clearing price (EUR/MWh) for hour t in zone $z \in \{\text{DE-LU, ES}\}$. Given information $\mathcal{F}_{t_0}$ available at decision time t_0 , we produce, for each hour $t \in [t_0 + 1 \text{ d}, t_0 + 2 \text{ d}]$ in the evaluation window, three quantile forecasts $\hat{q}_{0.025}^z(t)$, $\hat{q}_{0.45}^z(t)$, $\hat{q}_{0.975}^z(t)$.

Forecast quality is scored by the pinball loss

$$\rho_q(y, \hat{q}) = \max\{q(y - \hat{q}), (q - 1)(y - \hat{q})\}, \quad q \in (0, 1), \quad (1)$$

evaluated at $q = 0.45$. The slight asymmetry $q < 0.5$ penalises *over*-prediction more heavily than under-prediction, consistent with a price-taker procuring electricity at the auction.

2 Data

All inputs are public, hourly, and aligned to UTC.

Source	Variables	Range	Notes
Energy-Charts (Fraunhofer ISE)	p_t^z , generation per technology	2024-01 – live	DE-LU and ES
Open-Meteo Archive	T_t, v_t, I_t	2024-01 – today	temp, wind, irradiance
Open-Meteo Forecast	T_t, v_t, I_t	next 16 days	ECMWF + DWD blend
Yahoo Finance	NG=F, KRBN ETF	daily, filled	gas / CO ₂ proxies

Table 1: Data sources. Generation totals are decomposed into W_t (wind), S_t (solar PV), L_t (load), G_t (fossil gas).

Prices are clipped to $[-500, 3000]$ EUR/MWh and a 1-hour grid is imposed by linear resampling. Up to three consecutive missing values are forward-filled; longer gaps are imputed by column median.

3 Feature engineering

For each zone and hour we construct 43 features. With $h(t)$, $d(t)$, $m(t)$ denoting hour-of-day, day-of-week, and month, all periodic encodings use the standard sine/cosine pair

$$\phi_P(t) = (\sin(2\pi k(t)/P), \cos(2\pi k(t)/P)), \quad (2)$$

for periods $P \in \{24, 7, 12\}$.

Price autoregression. Lags of the target, $\{p_{t-\tau}^z : \tau \in \{24, 48, 72, 120, 168\}\}$, plus a one-week momentum $p_{t-24}^z - p_{t-168}^z$ and rolling statistics over windows $W \in \{24, 48, 72, 168\}$,

$$\mu_W(t) = \frac{1}{W} \sum_{i=1}^W p_{t-i}^z, \quad \sigma_W(t) = \sqrt{\frac{1}{W-1} \sum_{i=1}^W (p_{t-i}^z - \mu_W(t))^2}, \quad (3)$$

together with min and the empirical 0.1-quantile over $W = 24, 48$ to capture price-floor behaviour.

Cross-zone leakage. DE-LU and ES are weakly coupled through their common weather macro-pattern and via cross-border interconnectors (FR transit). We include $p_{t-24}^{\bar{z}}$, $p_{t-48}^{\bar{z}}$ where $\bar{z}$ is the partner zone, which contributes $\sim 3\%$ of LightGBM gain on DE-LU.

Exogenous. Weather (T_t, v_t, I_t) , generation (W_t, S_t, L_t, G_t) , and

$$\sigma_t = \frac{S_t}{\max(L_t, 1)}, \quad \Delta I_t = I_t - I_{t-1}, \quad (4)$$

the *solar surplus ratio* and irradiance change. Fuel proxies enter as 1- and 7-day-lagged daily-resampled values; binary holiday flags for DE and ES come from the `holidays` package.

4 Short-term model (≤ 14 days)

For each zone z and quantile $q \in \{0.025, 0.45, 0.975\}$ we fit a gradient-boosted regressor $f_{\theta}^{z,q} : \mathbb{R}^{43} \rightarrow \mathbb{R}$ minimising the empirical pinball loss

$$\mathcal{L}_q(\theta) = \sum_t w_t \rho_q(p_t^z, f_{\theta}^{z,q}(x_t)). \quad (5)$$

We use LightGBM with `objective='quantile'`, `alpha = q`, 800 trees (early-stopped), `learning_rate = 0.04`, `num_leaves = 63`, $L_1 = 0.1$, $L_2 = 0.5$, and column/row subsampling at 0.8.

4.1 Time-decayed sample weights

Recent regimes are more informative than the 2022 gas crisis, so samples are exponentially down-weighted with half-life $H = 180$ days:

$$w_t = \frac{\tilde{w}_t}{\bar{w}}, \quad \tilde{w}_t = \exp\left(-\frac{\ln 2}{H} \Delta_t\right), \quad \Delta_t = \frac{t_{\max} - t}{86400 \text{ s}}. \quad (6)$$

Six-month-old samples thus enter at half the leverage of today's; two-year-old samples at $\sim 8\%$.

4.2 Validation and ensembling

The last $8 \times 168 = 1344$ hours are held out for early stopping with patience 60. The selected number of rounds K^* is then *re-fit on the full training set*; this exploits the additional 8 weeks of recent data without re-tuning. We repeat the procedure for $M = 5$ random seeds and average:

$$\hat{q}_{\text{ST}}^{z,q}(t) = \frac{1}{M} \sum_{m=1}^M f_{\theta_m}^{z,q}(x_t). \quad (7)$$

Total: 2 zones $\times$ 3 quantiles $\times$ 5 seeds = 30 LightGBM models per run.

4.3 Constructing exogenous features for the eval window

Lag features at t in the eval window simply read from history. Generation features (W_t, S_t, L_t, G_t) must be *forecast*. We use a month $\times$ hour climatological mean $\bar{X}_{m,h} = \mathbb{E}[X_t \mid m(t) = m, h(t) = h]$ as a baseline, and rescale wind/solar by the realized weather-forecast ratio

$$\hat{S}_t = \bar{S}_{m(t),h(t)} \cdot \text{clip}\left(\frac{I_t}{\bar{I}_{m(t),h(t)}}, 0, 3\right), \quad \hat{W}_t = \bar{W}_{m(t),h(t)} \cdot \text{clip}\left(\frac{v_t}{\bar{v}_{m(t),h(t)}} , 0, 3\right), \quad (8)$$

preserving variance from the meteorological forecast while anchoring levels to seasonal climatology. Renewable share is then recomputed from the adjusted $\hat{S} + \hat{W}$ over the (climatological) load.

5 Long-term model (> 14 days)

Beyond two weeks the lag-driven short-term model has no signal: p_{t-168} becomes irrelevant. We use seasonal-trend decomposition (STL, robust, weekly period $P = 168$):

$$p_t^z = T_t + S_t + R_t, \quad (9)$$

fitted on the full training history per zone.

Trend extrapolation. The trend is linearly extrapolated from the last 4 weeks ($N = 28 \times 24 = 672$ hours):

$$\hat{T}_{t_0+h} = T_{t_0} + \alpha h, \quad \alpha = \arg \min_{\beta} \sum_{i=t_0-N+1}^{t_0} (T_i - (a + \beta i))^2. \quad (10)$$

Seasonal anchor. We collapse the STL seasonal to a month $\times$ hour climatology $\bar{S}_{m,h}$ and read off $\hat{S}_t = \bar{S}_{m(t),h(t)}$.

Probabilistic intervals. Empirical 2.5% and 97.5% residual quantiles $r_{0.025}, r_{0.975}$ are widened with horizon (in days, h_d):

$$\hat{q}_{\text{LT}}^{z,p}(t) = \hat{T}_t + \hat{S}_t + r_p \cdot \sqrt{\max(h_d/14, 1)}, \quad p \in \{0.025, 0.975\}, \quad (11)$$

with $\hat{q}_{0.45}^{\text{LT}}(t) = \hat{T}_t + \hat{S}_t$. The $\sqrt{\cdot}$ scaling is the diffusive widening expected from a random-walk residual.

6 Composition and post-processing

6.1 Regime switching

Let $h_d(t)$ denote the lead time in days from t_0 . The combined forecast is

$$\hat{q}^{z,p}(t) = \begin{cases} \hat{q}_{\text{ST}}^{z,p}(t) & \text{if } h_d(t) \leq 14, \\ \hat{q}_{\text{LT}}^{z,p}(t) & \text{if } h_d(t) > 14. \end{cases} \quad (12)$$

For the May 11 evaluation window all hours fall in the short-term regime; the long-term branch covers extended-horizon submissions.

6.2 Solar-driven interval widening

Backtest analysis showed the LightGBM model under-estimates uncertainty in high-irradiance midday hours, when negative-price events caused by photovoltaic surplus dominate. We multiply the half-widths by

$$\phi(I_t) = \min\left(1 + \frac{\max(I_t - 150, 0)}{400}, 2.0\right), \quad (13)$$

a piecewise-linear ramp from 1 at $I = 150$ W/m² to a cap of 2 at ~ 550 W/m², applied as

$$\hat{q}^{z,0.025'}(t) = \hat{q}^{z,0.45}(t) - \phi(I_t)(\hat{q}^{z,0.45}(t) - \hat{q}^{z,0.025}(t)), \quad (14)$$

$$\hat{q}^{z,0.975'}(t) = \hat{q}^{z,0.45}(t) + \phi(I_t)(\hat{q}^{z,0.975}(t) - \hat{q}^{z,0.45}(t)). \quad (15)$$

6.3 Quantile sorting

Independent fitting per quantile can produce crossings $\hat{q}_{0.025} > \hat{q}_{0.45}$. We enforce monotonicity by sorting the triplet:

$$(\hat{q}'_{0.025}, \hat{q}'_{0.45}, \hat{q}'_{0.975}) := \text{sort}(\hat{q}_{0.025}, \hat{q}_{0.45}, \hat{q}_{0.975}). \quad (16)$$

7 Backtest

We re-train end-to-end on data through 2026-05-09 and predict the next 24 hours (Sunday 2026-05-10), comparing against the realised day-ahead auction. Of 24 target hours, 22 have available actuals.

Metric	DE-LU	ES
Pinball loss $q = 0.45$	12.59	14.86
Mean absolute error (EUR/MWh)	23.36	27.02
Root-mean-square error (EUR/MWh)	30.27	32.47
Bias ($\hat{p}_{0.45} - p$) (EUR/MWh)	+18.25	+27.02
95% PI coverage	90.9%	63.6%
Validation pinball $q = 0.45$ (last 8 wk)	9.98	4.84
Test/validation gap	$\times 1.26$	$\times 3.07$

Table 2: Backtest on 2026-05-10 (22/24 h available). Validation pinball is on the last 8 weeks of the training window prior to re-fitting on the full set.

Discussion. DE-LU generalises cleanly: the validation-to-test pinball gap is a modest $\times 1.26$, and PI coverage of 90.9% is within reach of the nominal 95% given a small held-out sample. The intervals correctly contain negative midday prices triggered by PV surplus.

ES exhibits a systematic overnight failure on this Sunday: hourly predictions of 54–73 EUR/MWh against realised values of 14–20 EUR/MWh from 02:00–08:00 CEST. The +27 EUR/MWh bias and 63.6% coverage on Spain are driven almost entirely by these overnight hours. Inspection shows two contributing factors:

1. **Lag-feature dominance.** The 24-h price lag references Saturday’s overnight ES prices, which were structurally higher; the model up-weights this signal over the (weaker) day-of-week cyclical encoding.
2. **Solar-only widening.** The post-processing in Eq. (11) widens intervals only where $I_t > 150$, which is zero overnight. The model thus reports tight bands around a biased centre.

8 Limitations and future work

Day-of-week recalibration. The pinball asymmetry interacts poorly with weekend/holiday regime shifts. A targeted recalibration — e.g. a Sunday-specific bias correction $\hat{q}'_p(t) = \hat{q}_p(t) - b_{\text{Sun}}^z$ where b_{Sun}^z is the median test residual on prior Sundays — would directly attack the observed ES failure mode.

Coverage-driven widening. Replacing the manual solar widening with an isotonic/conformal recalibration on a holdout set would yield empirically valid 95% intervals across all regimes, not just sunny ones.

Joint zone modelling. Treating $(p_t^{\text{DE-LU}}, p_t^{\text{ES}})$ as a bivariate target with a shared latent renewables process should improve both bias and interval calibration during European-wide weather events. A multi-output LightGBM with chained quantile heads is a low-effort upgrade.

Reproducibility. All data is fetched from public APIs and cached locally; the run is seed-deterministic given fixed cache contents. End-to-end pipeline: ~ 4 min on a 2024 MacBook (data cached) for the full 30-model ensemble.